

Kerala Polytechnic Syllabus

# Electronic Instrumentation Lab

Course Code: 2089

Semester: 2

Revision: 2021

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## Course Overview

**Program:** Diploma in Instrumentation Engineering

**Category:** Engineering Science

**Credits:** No Credit (Lab Course)

**Total Periods:** 45 Hours

**Assessment:** CIE (Continuous Internal Evaluation) & ESE (End Semester Exam)

**Objective:** This lab introduces students to the practical aspects of measuring instruments, including bridge circuits, calibration of meters, and the use of modern diagnostic tools like DSO and Strip Chart Recorders.

## Course Outcomes (CO)

| CO No. | Description                                                                                       | Cognitive Level |
|--------|---------------------------------------------------------------------------------------------------|-----------------|
| CO1    | Determine static characteristics of instruments and convert galvanometers to ammeters/voltmeters. | Applying        |
| CO2    | Determine unknown R, L, C values using appropriate bridges (Wheatstone, Hay's, Schering).         | Applying        |
| CO3    | Calibrate Wattmeter and Energy meter using direct loading.                                        | Applying        |
| CO4    | Explain and utilize CRO/DSO and recorders for signal measurement.                                 | Understanding   |

# Module 1: Static Characteristics & Galvanometer Conversion

12 Hours

## M1.01: Static Characteristics & Calibration

Static characteristics are the properties of an instrument when the input is either constant or varying very slowly.

- **Accuracy:** The degree of closeness to the true value.
- **Precision:** The degree of repeatability of measurements.
- **Error:** The difference between measured value and true value. ( $\text{Error} = A_m - A_t$ )
- **Sensitivity:** The ratio of change in output to the change in input.
- **Hysteresis:** The difference in readings when a value is approached from an increasing direction versus a decreasing direction.

### Malayalam support:

ഒരു ഉപകരണം (Instrument) അളക്കുന്നതിനുള്ള അളവുകൾക്ക് അടുത്തുള്ളതായിരിക്കണം. Accuracy അളവുകൾ അളക്കുന്നതിനുള്ള. ഒരു ഉപകരണം അളക്കുന്നതിനുള്ള അളവുകൾക്ക് അടുത്തുള്ളതായിരിക്കണം. Precision അളവുകൾ അളക്കുന്നതിനുള്ള.

## M1.02: Galvanometer Conversion

A D'Arsonval galvanometer can be converted into an Ammeter or Voltmeter by adding specific resistances.

### 1. Conversion to Ammeter

A low resistance called **Shunt** ( $R_s$ ) is connected in parallel with the galvanometer.

$$R_s = R_g / (n - 1)$$

Where  $n = I / I_g$  (Multiplying factor),  $I$  is the total current to be measured, and  $I_g$  is full-scale deflection current.

## 2. Conversion to Voltmeter

A high resistance called **Multiplier ( $R_m$ )** is connected in series with the galvanometer.

$$R_m = R_g (n - 1)$$

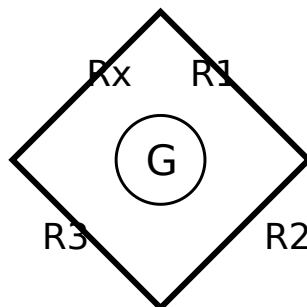
Where  $n = V / V_g$  (Multiplying factor),  $V$  is the total voltage to be measured.

# Module 2: Measurement using Bridges

8 Hours

## M2.01: Wheatstone's Bridge (Resistance)

Used for measuring medium resistances ( $1\Omega$  to  $100k\Omega$ ). It consists of four arms, a DC source, and a galvanometer.



Wheatstone Bridge Circuit Diagram

**Balance Condition:**  $R_x = (R_2 / R_1) \times R_3$

## M2.02: Hay's Bridge (Inductance)

Specifically used for measuring high-Q inductors ( $Q > 10$ ). It differs from Maxwell's bridge as the capacitor and resistor are in series.

### Malayalam support:

High Quality factor (Q)  $\square\square\square\square$   $\square\square\square\square\square\square\square\square\square\square$   $\square\square\square\square\square\square\square\square$  Hay's Bridge  $\square\square\square\square\square\square\square\square\square\square$ .

## M2.03: Schering Bridge (Capacitance)

Used for measuring unknown capacitance, dielectric loss, and power factor. It is one of the most important AC bridges.

# Module 3: Calibration of Wattmeter & Energy Meter

8 Hours

## Direct Loading Method

In this method, the meter is connected to a real load (like a lamp bank or rheostat). The reading of the meter under test is compared with a standard (calibrated) meter.

### Percentage Error Calculation:

% Error =  $[(\text{Measured Value} - \text{True Value}) / \text{True Value}] \times 100$

Energy meter 0000000000 0000000000 00000 (Revolutions) 0000000000  
0000000000 000000000000 00000000000. 0000000 00000000000 0000000  
00000000000 0000 00000000000.

14 Hours

**CRO (Cathode Ray Oscilloscope)** and **DSO (Digital Storage Oscilloscope)** are used to visualize waveforms.

- ## Lissajous Patterns

Used for frequency and phase measurement by applying two different signals to the X and Y plates.

| Pattern Shape            | Phase Difference ( $\phi$ ) |
|--------------------------|-----------------------------|
| Straight Line (Diagonal) | 0° or 360°                  |
| Circle                   | 90° or 270°                 |
| Ellipse                  | Between 0° and 90°          |

## M4.04: Strip Chart Recorder

A device that records a continuous visual record of a signal on a moving strip of paper.

- **Drive Mechanism:** Moves the paper at a constant speed.
- **Stylus/Pen:** Moves across the paper proportional to the input signal.

## Formula Bank

Ammeter Shunt

$$R_s = R_g / ( I/I_g - 1 )$$

Voltmeter Multiplier

$$R_m = R_g * ( V/V_g - 1 )$$

Wheatstone Balance

$$R_x = (R_2 * R_3) / R_1$$

Frequency (CRO)

$$f = 1 / (Div \times Time/Div)$$

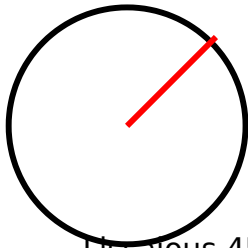
### Ammeter Shunt Calculator

Galvanometer Resistance ( $R_g$ )  $\Omega$ :

Full Scale Current ( $I_g$ ) mA:

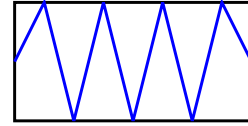
Desired Range ( $I$ ) A:

## Visual Library



Lissajous 45°

Phase Measurement



Strip Chart Trace

## Practice Questions

### 1. Define Hysteresis in instruments.

Hysteresis is the maximum difference in output readings for the same input, when the input is approached from opposite directions (increasing and decreasing).

### 2. Why is a shunt used in an ammeter?

A shunt is used to bypass the majority of the current around the galvanometer coil, allowing the meter to measure currents much larger than its full-scale deflection current without damaging the coil.

### 3. Which bridge is best for measuring high-Q inductance?

Hay's Bridge is preferred for high-Q inductors ( $Q > 10$ ).

## Practice Model Paper

## END SEMESTER EXAMINATION - PRACTICE

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### Part A: Short Answer (Select any 2)

1. Explain the difference between Accuracy and Precision.
2. State the balance condition for a Wheatstone bridge.
3. What is a Lissajous pattern?

### Part B: Practical/Long Answer

1. Describe the procedure to convert a 1mA, 100 $\Omega$  galvanometer into a 0-10A ammeter. Calculate the shunt resistance.
2. Explain the calibration of a single-phase energy meter using direct loading method with a neat circuit diagram.

## Quick Revision

- **Static Characteristics:** Accuracy, Precision, Sensitivity, Hysteresis, Resolution.
- **Ammeter:** Low resistance shunt in parallel.
- **Voltmeter:** High resistance multiplier in series.
- **Wheatstone Bridge:** Measures medium resistance.
- **Schering Bridge:** Measures capacitance.
- **Hay's Bridge:** Measures high-Q inductance.
- **CRO:** Measures Voltage, Time, Frequency, and Phase.

## Glossary

### DSO

Digital Storage Oscilloscope - An oscilloscope that digitizes and stores the input signal.

### Loading Effect

The distortion of the signal being measured caused by the act of connecting the measuring instrument.

### Q-Factor

Quality factor; a dimensionless parameter that describes how underdamped an oscillator or resonator is.

## Official Source Declaration



Source URL: [SITTTR Syllabus Website](#)

## പ്രാഥമിക പരീക്ഷണങ്ങൾ (Malayalam Support)

**പ്രധാന വിഷയങ്ങൾ:**

- Bridges:**  
കാർട്ടർ ബ്രിഡ്ജ്, വൈൻസ്ലോ ബ്രിഡ്ജ്, ക്ലൈവ് ബ്രിഡ്ജ് തുടങ്ങിയവയുടെ പ്രവർത്തനം പരീക്ഷിക്കുക. തെറ്റായ ബ്രിഡ്ജിംഗ് സാഹചര്യങ്ങൾ തിരിച്ചറിയുക.
- Calibration:**  
വോൾട്ട്, കറന്റ്, റെസിസ്റ്റൻസ് തുടങ്ങിയവയുടെ അളവുകൾ കൃത്യമാക്കുക. അളവുകൾ തെറ്റായതിനുള്ള കാരണങ്ങൾ തിരിച്ചറിയുക.
- CRO/DSO:**  
വോൾട്ട്, കറന്റ് തുടങ്ങിയവയുടെ തരംഗരൂപങ്ങൾ പരീക്ഷിക്കുക. തരംഗരൂപങ്ങൾ, ഫ്രീക്വൻസി, അംഗീകൃത വോൾട്ടേജ് തുടങ്ങിയവ.